
Factors, Multiples, Primes, and Divisibility — Grade 5 Answer Key

Solutions are keyed to the student workbook's page and problem IDs. Equivalent methods are acceptable when the reasoning and result are correct.

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- **P01-01:** Factor pairs: 1×36 , 2×18 , 3×12 , 4×9 , 6×6 .
- **P01-02:** Factor pairs: 1×48 , 2×24 , 3×16 , 4×12 , 6×8 . There are 10 positive factors: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.

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- **P02-01:** 1, 2, 3, 4, 6, 8, 12, 24.
- **P02-02:** 1, 2, 3, 6, 9, 18, 27, 54.

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- **P03-01:** 9, 18, 27, 36, 45, 54, 63, 72.
- **P03-02:** Yes. $7 \times 12 = 84$.

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- **P04-01:** Factors of 18 and 30 overlap at 1, 2, 3, and 6.
- **P04-02:** Factors overlap at 1, 2, 7, and 14.

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- **P05-01:** $\text{GCF}(24, 36) = 12$. For example, common factors are 1, 2, 3, 4, 6, 12; or use $24 = 2 \times 2 \times 2 \times 3$ and $36 = 2 \times 2 \times 3 \times 3$.
- **P05-02:** $\text{GCF}(45, 60) = 15$ kits. Each kit has $45 \div 15 = 3$ pencils and $60 \div 15 = 4$ erasers.

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- **P06-01:** 1,240 is divisible by 2, 5, and 10. 1,275 is divisible by 5 only. 1,286 is divisible by 2 only.
- **P06-02:** 1,000. It must end in 0 to be divisible by both 2 and 5; 1,000 is the least four-digit number ending in 0.

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- **P07-01:** Digit sum $4+5+7+2 = 18$. It is divisible by both 3 and 9.
- **P07-02:** For $63[\quad]$, the digit sum is $9 + [\quad]$. It is divisible by 9 when that sum is 9 or 18, so the possible digits are **0 or 9**.

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- **P08-01:** Yes. The last two digits are 16, and $16 \div 4 = 4$.
- **P08-02:** Yes. The last three digits are 248, and $248 \div 8 = 31$.

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- **P09-01:** 1 is neither; 17 and 29 are prime; 21 and 35 are composite.
- **P09-02:** 41, 43, 47, and 53. Each has no factor among 2, 3, 5, or any prime no greater than its square root.

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- **P10-01:** $72 = 2 \times 2 \times 2 \times 3 = 2^3 \times 3$.
- **P10-02:** $210 = 2 \times 3 \times 5 \times 7$.

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- **P11-01:** $84 = 2 \times 42 = 2 \times 2 \times 21 = 2^2 \times 3 \times 7$.
- **P11-02:** $90 = 9 \times 10 = 3 \times 3 \times 2 \times 5 = 2 \times 3^2 \times 5$. Prime factorization is unique, so a different tree reaches the same primes.

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- **P12-01:** Multiples of 6: 6, 12, 18, 24; of 8: 8, 16, 24. LCM = **24**.
- **P12-02:** $12 = 2^2 \times 3$ and $15 = 3 \times 5$, so LCM = $2^2 \times 3 \times 5 = 60$.

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- **P13-01:** LCM(4, 6) = 12, so they flash together after **12 seconds**.
- **P13-02:** LCM(10, 15) = 30 minutes; the next arrival is **8:30 a.m.**

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- **P14-01:** Possible arrays include 1×24 , 2×12 , 3×8 , and 4×6 (and their turned versions).
- **P14-02:** Side-length pairs are **1×48 , 2×24 , 3×16 , 4×12 , and 6×8** .

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- **P15-01:** $[] = 13$, because $117 \div 9 = 13$; check $9 \times 13 = 117$.
- **P15-02:** The number is **32**. Its positive factors are 1, 2, 4, 8, 16, 32; factor pairs are 1×32 , 2×16 , and 4×8 .

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- **P16-01:** 105, 120. They are multiples of 15 between 100 and 130.
- **P16-02:** 18, 54, and 90 are the possibilities if “two-digit” is enforced: each is divisible by 4 and has digits summing to 9. (90 is not divisible by 4, so remove it.) Final answers: **18 and 54**.

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- **P17-01:** $\text{GCF}(32, 48) = 16$. There are **16 teams**, with 2 students and 3 parents per team.
- **P17-02:** $\text{GCF}(72, 90) = 18$. There are **18 boxes**, with 4 chocolate and 5 vanilla cookies per box.

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- **P18-01:** $157 \div 12 = 13 \text{ remainder } 1$; $12 \times 13 + 1 = 157$.
- **P18-02:** $98 \div 8 = 12 \text{ remainder } 2$, so **13 vans** are needed; the two remaining children require another van.

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- **P19-01:** Common factors of 18 and 24 are 1, 2, 3, and 6, so $\text{GCF} = 6$. The first common multiple is 72, so $\text{LCM} = 72$.
- **P19-02:** Divisible by 2 (even), 3 (digit sum 9), 4 (last two digits 08), 8 (last three digits 008), and 9 (digit sum 9). Not divisible by 5 or 10 because it does not end in 0 or 5.

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- **P20-01:** For three distinct primes with 24 factors, exponents must give $(2+1)(1+1)(1+1) = 12$ or $(3+1)(2+1)(1+1) = 24$. The smallest is $2^3 \times 3^2 \times 5 = \mathbf{360}$, which has $(3+1)(2+1)(1+1) = 24$ factors.
- **P20-02:** False: 2 is even and prime. Correct statement: **Every even number greater than 2 is composite**, because it has factors 1, 2, and itself.